Q.P Code: D133162	Total Pages: 2	Name
		Register No.

FIRST SEMESTER UG DEGREE EXAMINATION, NOVEMBER 2025

(CUFYUGP)

B. Sc. Mathematics Honours

MAT1MN102 - CACULUS OF A SINGLE VARIABLE

2024 Admission onwards

Maximum Time :2 Hours	**Maximum Marks :70**

Section A

All Question can be answered. Each Question carries 3 marks (Ceiling : 24 Marks)

1	Determine whether the statement "If $\lim_{x\to a^+} f(x) = +\infty$, then $f(a)$ is undefined" is true or false. Explain your answer.
2	Find $\lim_{x\to 0} \frac{\cot x}{\text{cosec} x}$
3	During the first 40 s of a rocket flight, the rocket is propelled straight up so that in t seconds it reaches a height of $s = 0.3t^3$ ft. What is the average velocity of the rocket during the first 40 s?
4	Show that $\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$
5	Find an equation of the tangent line to the graph of $y = f(x)$ at $x = 2$ if $f(2) = -2$ and $f'(2) = -1$.
6	Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 25$
7	Find $\frac{dy}{dx}$ when $y = (\ln x)^{\tan x}$
8	Explain an increasing function with an example
9	Find the relative minimum of the function $\|x\|$
10	Explain the term relative extremum

Section B

All Question can be answered. Each Question carries 6 marks (Ceiling : 36 Marks))

11	Evaluate $\lim_{x\to+\infty} \sqrt{x^6+5} - x^3$
12	Discuss the continuity of the function $\sqrt{4x-3}$
13	Find $\frac{d^2y}{dx^2}$, if $y = \frac{3x-2}{5x}$
14	Show that $y = A\sin x + B\cos x$ is a solution of the equation $y'' + y = 0$ for all constants A and B.
15	Find the slopes of the tangent lines to the curve $y^2 - x + 1 = 0$ at the points (2, -1) and (2, 1).
16	Evaluate $\lim_{x\to0^+} \frac{1}{x(\ln x)^2}$ $; x \in (0,1)$
17	Find the intervals on which $f(x) = x^3$ is increasing and the intervals on which it is decreasing.
18	Find the relative extrema of $f(x) = x^4 - 4x^3 + 4x^2$

Section C

Answer any ONE. Each Question carries 10 marks (1x10=10 Marks))

19	1. Show that the equation $x^3 + x^2 - 2x = 1$ has at least one solution in the interval $[-1, 1]$. 2. Find value of the constant k , if possible, that will make the function f continuous everywhere. $$f(x) = \begin{cases} kx^2 & \text{if } x > 2 \\ 2x + k & \text{if } -x \le 2 \end{cases}$$
20	1. Show that $f(x) = x^3 - 3x^2 + 2x$ is not one-to-one on $(-\infty, \infty)$. 2. Find the largest value of k such that f is one-to-one on the interval $(-k, k)$.